

1. Understand the ZCU670's 8x8 MIMO Setup

Since the client emphasized "8 MIMO means separate," be clear on how the ZCU670 supports this:

- a. Separate Antennas:** The ZCU670 evaluation board has 8 TX antennas (via 8 RF-DACs at 10 GSPS) and 8 RX antennas (via 8 RF-ADCs at 2.95 GSPS), connected to 16 SMA connectors (8 TX, 8 RX) for full-duplex 8x8 MIMO. No shared ports are used, ensuring isolation between transmit and receive paths.
- b. Frequency and Bandwidth:** Operates from 10 MHz to 7.125 GHz (FR1, sub-6 GHz) with up to 400 MHz instantaneous bandwidth per channel, totaling 3.2 GHz aggregate for 8x8 MIMO.
- c. O-RAN Relevance:** The ZCU670's hardened Digital Front-End (DFE) IP (DUC/DDC, CFR, DPD) supports O-RAN's low-PHY processing, compliant with 7.2x split and eCPRI fronthaul.
- d. Potential Question: "How does your setup handle the 8x8 MIMO requirement?"**

Answer Prep: "The ZCU670 uses 8 dedicated RF-DACs for TX and 8 RF-ADCs for RX, each connected to separate SMA connectors, supporting full 8x8 MIMO with 400 MHz per channel in the 10 MHz to 7.125 GHz range. We've configured the DFE IP for 5G NR beamforming and O-RAN compliance."

2. O-RAN Implementation Basics

Be ready to explain how the ZCU670 fits into the O-RAN architecture:

Role in O-RAN: The ZCU670 typically implements the Radio Unit (RU), handling low-PHY processing (e.g., FFT/iFFT, beamforming) and interfacing with the Distributed Unit (DU) via eCPRI over SFP+ ports.

3. FR2 (mmWave) Limitations

Since we discussed that the ZCU670 doesn't natively support FR2 (24.25–52.6 GHz), be prepared for questions about mmWave:

FR1 Support: Native operation in 10 MHz–7.125 GHz covers all FR1 5G bands.

FR2 Workaround: Requires external up/down converters to translate mmWave signals to an intermediate frequency (IF) within the ZCU670's range.

4. Common Technical Questions to Anticipate

Here are likely questions from the client or third-party, with concise responses:

- **Q:** How do you ensure MIMO synchronization?

A: "The ZCU670 uses multi-tile synchronization to align all 8 TX and 8 RX channels, ensuring phase coherence for beamforming. We use an external clock or the onboard LMK04828 for precise timing."

- **Q:** What's the bandwidth capability for O-RAN?**

A: "Each of the 8 TX and 8 RX channels supports up to 400 MHz, giving a total of 3.2 GHz aggregate bandwidth for 8x8 MIMO, ideal for 5G NR."

- **Q:** How do you interface with other O-RAN components?**

A: "We use eCPRI over SFP+ for fronthaul to the DU and JESD204C for RF front-end integration, following O-RAN standards for interoperability."

- **Q:** What tools are you using for development?**

A: "We're using Vivado ML for FPGA design and Vitis for software, leveraging AMD's O-RAN TRD and RF Analyzer GUI for testing."

About IPs use in RFsoc ZCU670

Question: "What licenses do we need for the RFSoc IPs on ZCU670?"

- **Answer:** "The core RFSoc IPs, like DUC/DDC, DPD, CFR, and Low-PHY, are hardened in the ZCU670's XCZU67DR chip and don't need separate licenses. The RF Data Converter and Ethernet IPs are included in the Vivado ML Enterprise Edition, which comes with the ZCU670 kit for evaluation. If we're using third-party IPs, like Renesas for clocking or iCana for RF front-ends, those licenses are tied to their hardware or reference designs."

Question: "Are there additional costs for O-RAN IPs?"

- **Answer:** "AMD's O-RAN Targeted Reference Design for the ZCU670 includes all necessary IPs for 5G NR and eCPRI fronthaul, covered by the Vivado ML license included"

with the kit. Third-party IPs, if used, would depend on specific vendor terms, like iCana's reference design license."

Question: "How do we access these IPs in Vivado?"

- **Answer:** "We use Vivado ML Enterprise Edition, installed with the node-locked license provided with the ZCU670 kit. The RFDC and Ethernet IPs are in the Vivado IP Catalog, and the O-RAN TRD provides pre-integrated designs for quick setup."